

2015

Assessment Task 4  
Trial HSC Examination

# Mathematics Extension 1

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## General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using a blue or black pen.
- Board approved calculators and mathematical templates and instruments may be used.
- Show all necessary working in Questions 11, 12, 13 and 14
- This examination booklet consists of 13 pages including a standard integral page and a multiple choice answer sheet.

## Total marks (70)

### Section I

#### Total marks (10)

- Attempt Questions 1 – 10
- Answer on the Multiple Choice answer sheet provided on the last page of this question booklet.
- Allow about 15 minutes for this section

### Section II

#### Total marks (60)

- Attempt questions 11 to 14
- Answer each question in the writing booklets provided.
- Start a new booklet for each question with your student name and question number at the top of the page.
- All necessary working should be shown for every question
- Allow about 1 hour 45 minutes for this section

Student Name : \_\_\_\_\_

Teacher : \_\_\_\_\_

## Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1 – 10.

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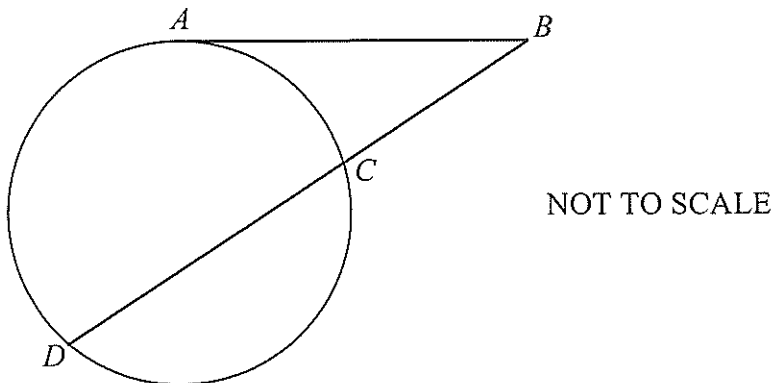
1 The solution to the inequality  $x(2-x)(x+1) \geq 0$  is

- (A)  $x \leq -2$  or  $0 \leq x \leq 1$       (B)  $-2 \leq x \leq 0$  or  $x \geq 1$   
(C)  $x \leq -1$  or  $0 \leq x \leq 2$       (D)  $-1 \leq x \leq 0$  or  $x \geq 2$

2 A committee of 3 men and 3 women is to be formed from a group of 8 men and 6 women. How many ways can this be done?

- (A) 48      (B) 1120  
(C) 40320      (D) 3003

3



In the diagram,  $AB$  is a tangent to the circle,  $BC = 6\text{cm}$  and  $CD = 12\text{cm}$ . What is the length of  $AB$ ?

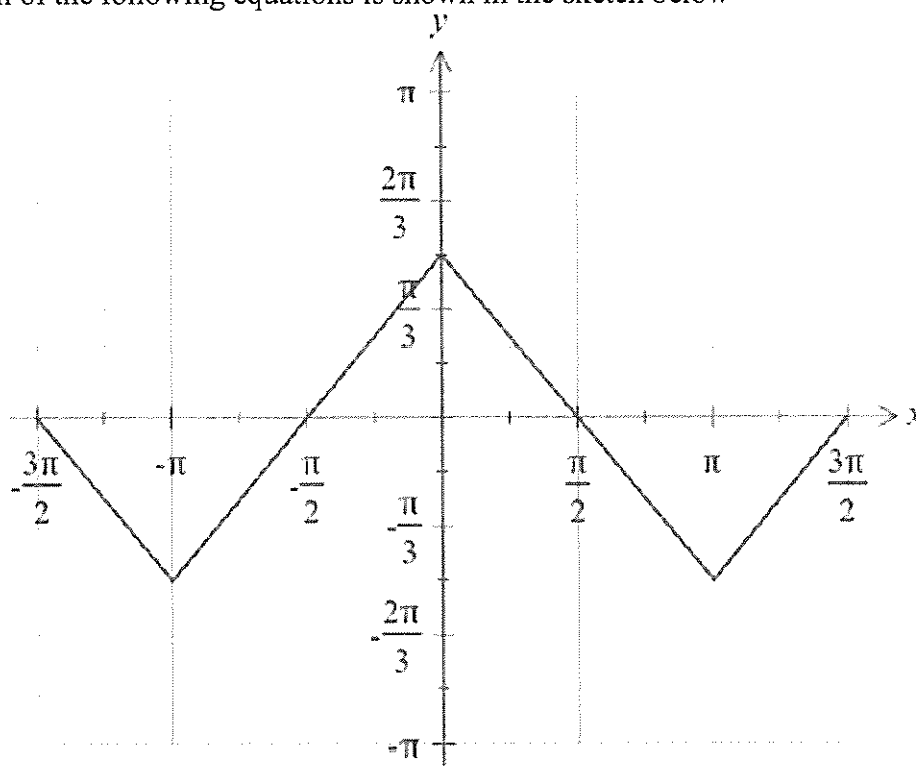
- (A)  $6\sqrt{2}\text{ cm}$       (B)  $6\sqrt{3}\text{ cm}$   
(C)  $72\text{cm}$       (D)  $108\text{cm}$

4 What is the equation of the tangent at the point  $(4p, 2p^2)$  on the parabola  $x^2 = 8y$ ?

- (A)  $y = px - p^2$       (B)  $x + py = 2p + p^3$   
(C)  $x + py = 4p + p^3$       (D)  $y = px - 2p^2$

- 5 What is the acute angle to the nearest degree that the line  $2x - 3y + 5 = 0$  makes with the  $y$ -axis?
- (A)  $27^\circ$  (B)  $34^\circ$   
(C)  $56^\circ$  (D)  $63^\circ$
- 6 Which of the following statements is FALSE.
- (A)  $\cos^{-1}(-\theta) = -\cos^{-1} \theta$  (B)  $\sin^{-1}(-\theta) = -\sin^{-1} \theta$   
(C)  $\tan^{-1}(-\theta) = -\tan^{-1} \theta$  (D)  $\cos^{-1}(-\theta) = \pi - \cos^{-1} \theta$
- 7 The primitive of  $2x(3x^2 - 1)^4$  is:
- (A)  $\frac{1}{15}(3x^2 - 1)^5 + c$  (B)  $\frac{3}{5}(3x^2 - 1)^5 + c$   
(C)  $\frac{2x}{5}(3x^2 - 1)^5 + c$  (D)  $\frac{2x}{15}(3x^2 - 1)^5 + c$
- 8 The equation(s) of the horizontal asymptote(s) to the curve  $y = \frac{x^2 + 1}{x^2 - 1}$  are
- (A)  $y = 0$  (B)  $x = \pm 1$   
(C)  $y = 1$  (D)  $x = 1$  only
- 9 What are the coordinates of the point that divides the interval joining the points  $A(2, 2)$  and  $B(4, 5)$  externally in the ratio 2:3?
- (A)  $(-2, -4)$  (B)  $(-2, 11)$   
(C)  $(8, -4)$  (D)  $(8, 11)$

- 10 Which of the following equations is shown in the sketch below



- (A)  $y = \cos^{-1}(\sin x)$       (B)  $y = \sin^{-1}(\cos x)$   
 (C)  $y = \sin^{-1}(x) + \sin(x)$       (D)  $y = \cos^{-1}(x) + \cos(x)$

*~ End of Section I ~*

## Section II

60 marks

Attempt Questions 11 to 14

Allow about 1 hour 45 minutes for this section

Answer each question in the appropriate writing booklet.

All necessary working should be shown in every question.

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| Question 11 (15 marks) |                                                                                                                                                                                                                           | Marks |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| (a)                    | Solve the inequality $\frac{3}{x(2x-1)} > 1$ ?                                                                                                                                                                            | 3     |
| (b)                    | In what ratio does the point (14,18) divide the interval joining $X(-1,3)$ to $Y(4,8)$ ?                                                                                                                                  | 2     |
| (c)                    | (i) Show that the curves $y = x^3 - x$ and $y = x - x^2$ intersect at the point $(-2,-6)$                                                                                                                                 | 1     |
|                        | (ii) Determine the acute angle between the curves $y = x^3 - x$ and $y = x - x^2$ at the point of intersection, to the nearest minute.                                                                                    | 3     |
| (d)                    | (i) A class of 25 students is to be divided into four groups consisting of 3, 4, 5 and 6 students. How many ways can this be done? Leave your answer in unsimplified form.                                                | 2     |
|                        | (ii) Assume that the four groups have been chosen. How many ways can the 25 students be arranged around a circular table if the students in each group are to be seated together? Leave your answer in unsimplified form. | 2     |
| (e)                    | Five different fair dice are thrown together. What is the probability the five scores are all different?                                                                                                                  | 2     |

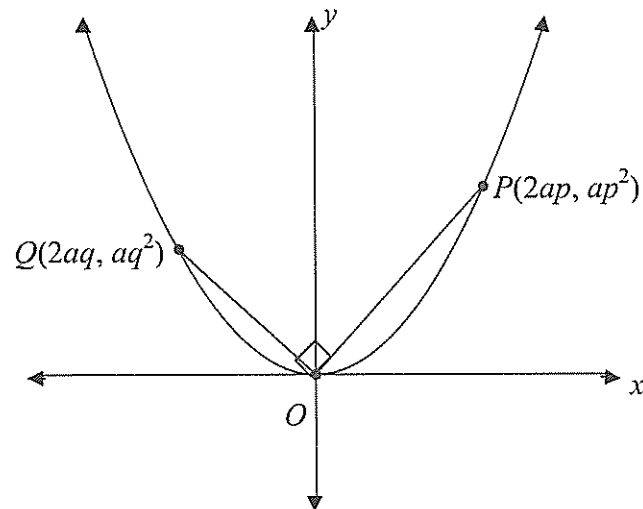
**Question 12** (15 marks)**Marks**

- (a) Consider the function  $f(x) = (x-1)^2$
- (i) Sketch  $y = f(x)$ . 1
- (ii) Explain why  $f(x)$  does not have an inverse function for all  $x$  in its domain. 1
- (iii) State a domain and range for which  $f(x)$  has an inverse function  $f^{-1}(x)$ . 1
- (iv) For  $x \geq 1$  find the equation of the function  $f^{-1}(x)$ . 2
- (v) Hence, on a new set of axes, sketch the graph of  $y = f^{-1}(x)$ . 1
- (b) Find  $\int \frac{dx}{\sqrt{9-4x^2}}$  2
- (c) Find the exact value of  $\tan\left(2 \tan^{-1} \frac{3}{4}\right)$  2
- (d) Find the general solution to  $2 \cos x = \sqrt{3}$ .  
Leave your answer in terms of  $\pi$ . 2
- (e) Differentiate (with respect to  $x$ )  
 $\left(\tan^{-1} \frac{x}{3}\right)^2$   
and hence find the exact value of  
 $\int_0^{\sqrt{3}} \frac{\tan^{-1} \frac{x}{3}}{x^2 + 9} dx$  3

**Question 13** (15 marks)

**Marks**

- (a) The points  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  lie on the parabola  $x^2 = 4ay$  such that  $OP$  is perpendicular to  $OQ$ .



- |       |                                                                                                                                     |   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------|---|
| (i)   | Prove that $pq = -4$ .                                                                                                              | 2 |
| (ii)  | $R$ is the point such that $OPRQ$ is a rectangle.<br>Explain why the co-ordinates of $R$ are $(2a(p+q), a(p^2+q^2))$ .              | 2 |
| (iii) | Show that the locus of $R$ is a parabola.                                                                                           | 2 |
| (b)   | Find by division of polynomials, the remainder when $x^2 + 4$ is divided by $x - 3$ .                                               | 1 |
| (c)   | $\alpha$ , $\beta$ and $\gamma$ are the roots of the equation $x^3 - 3x^2 - 6x - 1 = 0$ .<br>Find $\alpha^2 + \beta^2 + \gamma^2$ . | 2 |

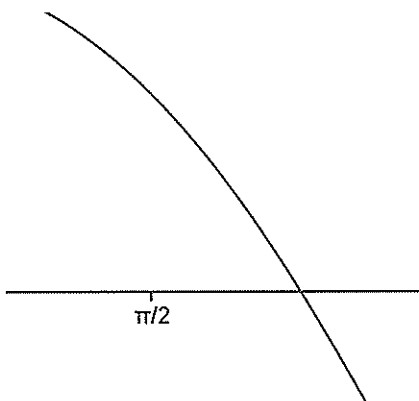
*Question 13 continued next page....*

Question 13 continued....

- (d) (i) Consider the curve  $f(x) = \sin^2 x - x + 1$  for  $0 \leq x \leq \pi$ .  
Show that it has one stationary point and determine its nature. 3

- (ii)  $f(x) = \sin^2 x - x + 1$  has a zero near  $x_1 = \frac{\pi}{2}$ .  
Use one application of Newton's method to obtain another approximation  $x_2$ , to this zero. 2

(iii)



The graph of  $f(x) = \sin^2 x - x + 1$  is shown in the vicinity of  $x = \frac{\pi}{2}$ .

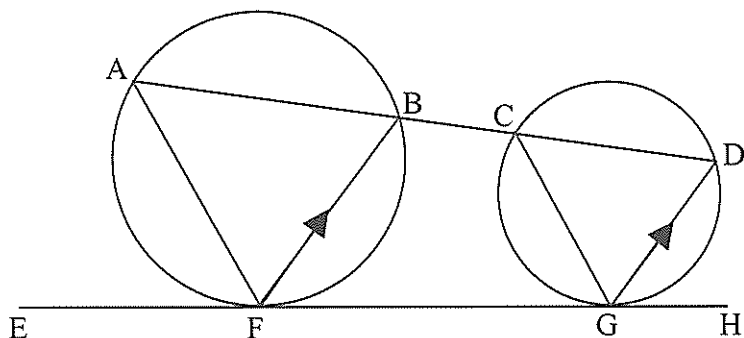
By using this diagram, determine if  $x_2$  is a better approximation than  $x_1$  to the real root of the equation. You must justify your answer. 1



**Question 14** (15 marks)

**Marks**

(a)



In the diagram above,  $FG$  is a common tangent and  $FB \parallel GD$ .

(i) Prove that  $FA \parallel GC$ . 2

(ii) Prove that  $BCGF$  is a cyclic quadrilateral. 2

(b) (i) Find:  $\frac{d}{dx}(x \sin 3x)$ . 2

(ii) Hence, evaluate:  $\int_0^{\frac{\pi}{6}} x \cos 3x \, dx$  3

(c) Use the substitution  $y = \sqrt{x}$  to find

$$\int \frac{dx}{\sqrt{x(1-x)}} \quad \text{3}$$

(d) Use mathematical induction to prove the inequality: 3

$n! > 2^n$ , for all positive integral values of  $n \geq 4$

*~ End of Section II ~*

Year 12 Mathematics Extension 1 Trial 2015

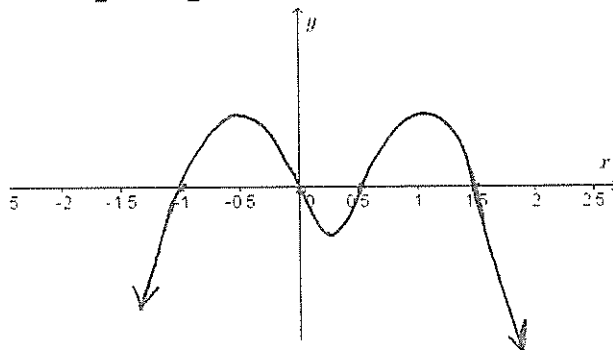
Question No. 11

Solutions and Marking Guidelines

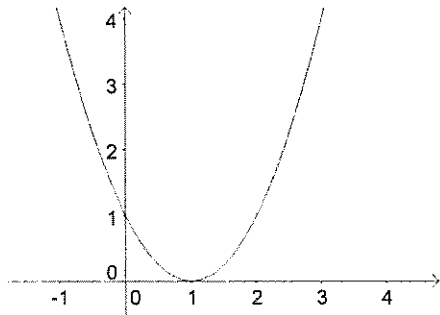
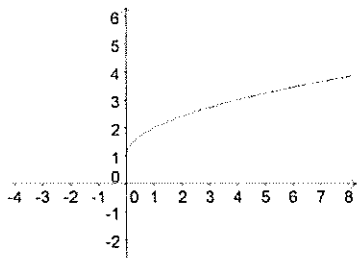
Outcomes Addressed in this Question

PE3 Solves problems involving permutations and combinations, inequalities and polynomials.

H5 Applies appropriate techniques from the study of geometry.

| Outcome | Solutions                                                                                                                                                                                                                                                                                                                                                                                   | Marking Guidelines                                                                                                                                            |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PE3     | <p>(a)</p> $\frac{3}{x(2x-1)} > 1$ <p>Multiply by the square of the denominator</p> $3x(2x-1) > x^2(2x-1)^2$ $3x(2x-1) - x^2(2x-1)^2 > 0$ $x(2x-1)(3 - x(2x-1)) > 0$ $x(2x-1)(-2x^2 + x + 3) > 0$ $-x(2x-1)(2x-3)(x+1) > 0$ $\therefore -1 < x < 0, \frac{1}{2} < x < \frac{3}{2}$                       | <p>3 marks<br/>Correct solution</p> <p>2 marks<br/>Substantial progress towards correct solution</p> <p>1 mark<br/>Some progress towards correct solution</p> |
| H5      | <p>(b)</p> <p><math>A(-1,3)</math> <math>B(4,8)</math> <math>P(14,18)</math></p> <p>Ratio <math>AB : BP</math>, <math>m : n</math> assume <math>1 : k</math></p> $x_0 = \frac{mx_2 + nx_1}{m + n}$ $14 = \frac{1(4) + k(-1)}{1 + k}$ $14 + 14k = 4 - k$ $k = \frac{-2}{3}$ $1 : k = 1 : \frac{-2}{3}$ <p><math>\therefore</math> Ratio is <math>-3 : 2</math></p>                           | <p>2 marks<br/>Correct solution.</p> <p>1 mark<br/>Substantial progress towards correct solution.</p>                                                         |
| H5      | <p>(c)(i)</p> <p>Substitute <math>x = -2</math> into <math>y = x^3 - x</math></p> $RHS = (-2)^3 - (-2)$ $= -6$ <p><math>\therefore</math> Satisfies the curve.</p> <p>Substitute <math>x = -2</math> into <math>y = x - x^2</math></p> $RHS = -2 - (-2)^2$ $= -6$ <p><math>\therefore</math> Satisfies the curve.</p> <p><math>\therefore (-2, -6)</math> is the point of intersection.</p> | <p>1 mark<br/>Correct solution</p>                                                                                                                            |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p><b>(c)(ii)</b><br/> For <math>y = x^3 - x</math>, <math>y' = 3x^2 - 1</math><br/> when <math>x = -2</math>, <math>m_1 = 3(-2)^2 - 1 = 11</math><br/> For <math>y = x - x^2</math>, <math>y' = 1 - 2x</math><br/> when <math>x = -2</math>, <math>m_2 = 1 - 2(-2) = 5</math><br/> <math display="block">\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}</math> <math display="block">\tan \theta = \frac{11 - 5}{1 + (11)(5)}</math> <math display="block">\therefore \theta = \tan^{-1} \left( \frac{6}{56} \right)</math> <math display="block">\therefore \theta = 6^\circ 7' \text{ (to the nearest minute)}</math></p> | <p><b>3 marks</b><br/> Correct solution with correct rounding<br/> <b>2 marks</b><br/> Substantial progress towards correct solution<br/> <b>1 mark</b><br/> Some progress towards correct solution</p> |
| PE3 | <p><b>(d) (i)</b><br/> <math>{}^{25}C_3 \times {}^{22}C_4 \times {}^{18}C_5 \times {}^{13}C_6</math><br/> or <math>\frac{25!}{3! 4! 5! 6! 7!}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p><b>2 marks</b><br/> Correct solution<br/> <b>1 mark</b><br/> Substantial progress towards correct solution</p>                                                                                       |
| PE3 | <p><b>(d) (ii)</b><br/> <math>(11-1)! \times 3! \times 4! \times 5! \times 6!</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p><b>2 marks</b><br/> Correct solution.<br/> <b>1 mark</b><br/> Substantial progress towards correct solution.</p>                                                                                     |
| H5  | <p><b>(e)</b><br/> <math display="block">P(E) = \frac{6 \times 5 \times 4 \times 3 \times 2}{6^5}</math> <math display="block">= \frac{5}{54}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p><b>2 marks</b><br/> Correct solution.<br/> <b>1 mark</b><br/> Substantial progress towards correct solution.</p>                                                                                     |

|                                                                                      |                                                                                                                                                                                                                                                    |                                                                                  |                  |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------|
| Year 12 2015                                                                         |                                                                                                                                                                                                                                                    | Mathematics Extension 1                                                          | Task 4 Trial HSC |
| Question No. 12                                                                      |                                                                                                                                                                                                                                                    | Solutions and Marking Guidelines                                                 |                  |
| Outcomes Addressed in this Question                                                  |                                                                                                                                                                                                                                                    |                                                                                  |                  |
| HE4 uses the relationship between functions, inverse functions and their derivatives |                                                                                                                                                                                                                                                    |                                                                                  |                  |
| Outcome                                                                              | Solutions                                                                                                                                                                                                                                          | Marking Guidelines                                                               |                  |
| HE4                                                                                  | Question 12<br>a)<br>(i)<br>                                                                                                                                      | 1 Mark for correct sketch                                                        |                  |
| HE4                                                                                  | (ii)<br>It does not have an inverse because for every $y$ value there is more than one $x$ value.<br>Or<br>Does not pass the horizontal line test.<br>Or<br>Anything that is equivalent.                                                           | 1 Mark for correct explanation                                                   |                  |
| HE4                                                                                  | (iii)<br>Domain: $x \geq 1$<br>Range: $y \geq 0$                                                                                                                                                                                                   | 1 Mark for correct answer                                                        |                  |
| HE4                                                                                  | (iv)<br>$x = (y - 1)^2$ $\sqrt{x} = y - 1$ $y = 1 + \sqrt{x}$ $\therefore f^{-1}(x) = 1 + \sqrt{x}$                                                                                                                                                | 2 Marks for complete correct solution<br><br>1 Mark for partial correct solution |                  |
| HE4                                                                                  | (v)<br>                                                                                                                                                         | 1 Mark for correct sketch                                                        |                  |
| HE4                                                                                  | (b)<br>$\int \frac{dx}{\sqrt{9 - 4x}}$ $= \int \frac{dx}{\sqrt{4\left(\frac{9}{4} - x\right)}}$ $= \frac{1}{2} \sin^{-1}\left(\frac{\frac{x}{3}}{\frac{3}{2}}\right) + C \quad \text{or} \quad \frac{1}{2} \sin^{-1}\left(\frac{2x}{3}\right) + C$ | 2 Marks for complete correct solution<br><br>1 Mark for partial correct solution |                  |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HE4 | <p>(c)</p> <p>let <math>\theta = \tan^{-1}\left(\frac{3}{4}\right) \therefore \tan \theta = \frac{3}{4}</math></p> <p>now,</p> $\begin{aligned}\tan\left(2 \tan^{-1} \frac{3}{4}\right) &= \tan(2\theta) \\ &= \frac{2 \tan \theta}{1 - \tan^2 \theta} \\ &= \frac{2\left(\frac{3}{4}\right)}{1 - \left(\frac{3}{4}\right)^2} \\ &= \frac{24}{7}\end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>2 Marks for complete correct solution</p> <p>1 Mark for partial correct solution</p>                                                                                                                                                                 |
| HE4 | <p>(d)</p> $2 \cos x = \sqrt{3}$ $\cos x = \frac{\sqrt{3}}{2}$ $\therefore x = 2n\pi \pm \cos^{-1} \frac{\sqrt{3}}{2}$ $= 2n\pi \pm \frac{\pi}{6}, \text{ where } n \text{ is any integer.}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>2 Marks for complete correct solution</p> <p>1 Mark for partial correct solution</p>                                                                                                                                                                 |
| HE4 | <p>(e)</p> $\begin{aligned}\frac{d}{dx}\left(\tan^{-1} \frac{x}{3}\right)^2 &= 2\left(\tan^{-1} \frac{x}{3}\right)\left(\frac{\frac{1}{3}}{1 + \frac{x^2}{9}}\right) \\ &= 2\left(\tan^{-1} \frac{x}{3}\right)\left(\frac{3}{9 + x^2}\right) \\ &= 6\left(\frac{\tan^{-1} \frac{x}{3}}{9 + x^2}\right)\end{aligned}$ <p>now,</p> $\begin{aligned}\int_0^{\sqrt{3}} \left(\frac{\tan^{-1} \frac{x}{3}}{9 + x^2}\right) dx &= \frac{1}{6} \int_0^{\sqrt{3}} 6\left(\frac{\tan^{-1} \frac{x}{3}}{9 + x^2}\right) dx \\ &= \frac{1}{6} \left[ \left(\tan^{-1} \frac{x}{3}\right)^2 \right]_0^{\sqrt{3}} \\ &= \frac{1}{6} \left( \left(\tan^{-1} \frac{\sqrt{3}}{3}\right)^2 - \left(\tan^{-1} \frac{0}{3}\right)^2 \right) \\ &= \frac{1}{6} \left( \left(\frac{\pi}{6}\right)^2 - 0 \right) \\ &= \frac{\pi^2}{216}\end{aligned}$ | <p>3 Marks for complete correct solution</p> <p>2 Mark for substantial working that could lead to a correct solution with only one error.</p> <p>1 Mark for correctly differentiating <math>\frac{d}{dx}\left(\tan^{-1} \frac{x}{3}\right)^2</math></p> |

|                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      |                  |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------|
| Year 12 Trial Higher School Certificate                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                | Extension 1 Mathematics                                                              | Examination 2015 |
| Question No. 13                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                | Solutions and Marking Guidelines                                                     |                  |
| Outcomes Addressed in this Question                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      |                  |
| PE3 solves problems involving polynomials and parametric representations                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      |                  |
| PE5 determines derivatives which require the application of more than one rule of differentiation |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      |                  |
| H6 uses the derivative to determine the features of the graph of a function                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      |                  |
| HE7 evaluates mathematical solutions to problems and communicates them in an appropriate form.    |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      |                  |
| Outcome                                                                                           | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marking Guidelines                                                                   |                  |
| PE3                                                                                               | (a) (i) $OP \perp OQ, \therefore mOP \times m OQ = -1.$<br>$\therefore \frac{ap^2}{2ap} \times \frac{aq^2}{2aq} = -1$<br>$\therefore \frac{p}{2} \times \frac{q}{2} = -1$<br>$\therefore pq = -4.$                                                                                                                                                                                                                                             | 2 marks : correct solution<br>1 mark : significant progress towards correct solution |                  |
| PE3                                                                                               | (ii) Midpoint $PQ = \left( ap + aq, \frac{ap^2 + aq^2}{2} \right)$<br>As the diagonals bisect one another in a rectangle, $OR$ will also have the same midpoint as $PQ$ .<br>If $O(0,0)$ , $R$ has midpoint $\left( ap + aq, \frac{ap^2 + aq^2}{2} \right)$ , then<br>$R$ is $\left( 2a(p + q), a(p^2 + q^2) \right).$                                                                                                                         | 2 marks : correct solution<br>1 mark : significant progress towards correct solution |                  |
| PE3                                                                                               | (iii) At $R, \begin{cases} pq = -4 & [1] \\ x = 2a(p + q) & [2] \\ y = a(p^2 + q^2) & [3] \end{cases}$<br>From $[2], x^2 = 4a^2(p + q)^2$<br>$\therefore x^2 = 4a^2(p^2 + q^2 + 2pq)$<br>Substituting $[1]$ and $[3], x^2 = 4a^2\left(\frac{y}{a} + 2 \times -4\right),$<br>$\therefore x^2 = 4a(y - 8a)$ , which is a concave up parabola with vertex $(0,8a).$                                                                               | 2 marks : correct solution<br>1 mark : significant progress towards correct solution |                  |
| PE3                                                                                               | (b)<br>$\begin{array}{r} x+3 \\ x-3 \overline{) x^2 + 0x + 4} - \\ \underline{x^2 - 3x} \phantom{4} \\ 3x + 4 - \\ \underline{3x - 9} \\ 13 \end{array}$ Remainder is 13.<br>(c) From $x^3 - 3x^2 - 6x - 1 = 0,$<br>$\alpha + \beta + \gamma = \frac{-b}{a} = 3, \alpha\beta + \beta\gamma + \alpha\lambda = \frac{c}{a} = -6.$<br>$(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \alpha\gamma)$ | 2 marks : correct solution<br>1 mark : significant progress towards correct solution |                  |
|                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2 marks : correct solution<br>1 mark : significant progress towards correct solution |                  |

H6, PE5

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = (3)^2 - 2(-6) \\ = 21.$$

(d) (i)  $f(x) = \sin^2 x - x + 1$

$$f'(x) = 2 \sin x \cos x - 1$$

$$\therefore f'(x) = \sin 2x - 1$$

$$f'(x) = 0 \text{ for stationary points.}$$

$$\text{Solving } \sin 2x - 1 = 0,$$

$$\sin 2x = 1$$

$$\text{For } 0 \leq x \leq \pi, \quad 0 \leq 2x \leq 2\pi.$$

$$\text{Solving, } 2x = \frac{\pi}{2}, \quad \therefore \text{one stationary point, at } x = \frac{\pi}{4}.$$

$$\text{Testing } x = \frac{\pi}{4}, \text{ for } f'(x) = \sin 2x - 1,$$

|         |                        |                 |                        |
|---------|------------------------|-----------------|------------------------|
| $x$     | $\frac{\pi}{6}$        | $\frac{\pi}{4}$ | $\frac{\pi}{3}$        |
| $f'(x)$ | $\frac{\sqrt{3}-2}{2}$ | 0               | $\frac{\sqrt{3}-2}{2}$ |

As  $\frac{\sqrt{3}-2}{2}$  is negative, there is a horizontal point of

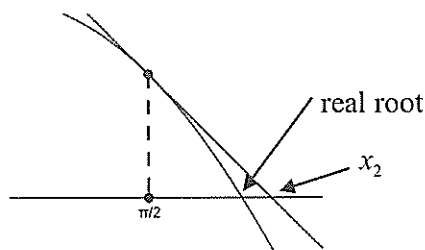
inflexion at  $x = \frac{\pi}{4}$ .

(ii)  $f\left(\frac{\pi}{2}\right) = \sin^2 \frac{\pi}{2} - \frac{\pi}{2} + 1 = 2 - \frac{\pi}{2}$

$$f'\left(\frac{\pi}{2}\right) = \sin \pi - 1 = -1.$$

$$\text{Newton's method : } x_2 = \frac{\pi}{2} - \frac{2 - \frac{\pi}{2}}{-1} \\ \therefore x_2 = 2$$

(iii)



$x_2$  is where the tangent at  $\frac{\pi}{2}$  meets the  $x$  axis.

This is closer to the real root,  $\therefore$  a better approximation.

PE5, HE7

HE7

3 marks : correct solution  
2 marks : substantial progress towards correct solution  
1 mark: significant progress towards correct solution

2 marks : correct solution  
1 mark : significant progress towards correct solution

1 mark : correct explanation

| Year 12 Mathematics Extension 1 Trial Examination 2015 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                      |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question No. 14                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Solutions and Marking Guidelines                                                                                                                                                                                     |
| Outcomes Addressed in this Question                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                      |
| <b>PE3</b>                                             | solves problems involving circle geometry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                      |
| <b>HE2</b>                                             | uses inductive reasoning in the construction of proofs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                      |
| <b>HE4</b>                                             | uses the relationship between functions, inverse functions and their derivatives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                      |
| <b>HE6</b>                                             | determines integrals by reduction to a standard form through a given substitution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                      |
| Outcome                                                | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marking Guidelines                                                                                                                                                                                                   |
| <b>PE3</b>                                             | <p>(a)(i) Let <math>\angle DGH = \alpha</math><br/> <math>\therefore \angle BFG = \alpha</math> (corresponding angles, <math>FB \parallel GD</math>)<br/> Now, <math>\angle GCD = \alpha</math> (angle between a chord and tangent is equal to the angle in the alternate segment)</p> <p>Similarly, <math>\angle FAB = \alpha</math><br/> Since <math>\angle GCD = \angle FAB = \alpha</math>,<br/> <math>FA \parallel GC</math> (corresponding angles are equal)</p>                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>2 marks</b><br/> Correct solution with full reasoning.<br/> <b>1 mark</b><br/> Substantial progress towards a correct solution.</p>                                                                            |
| <b>PE3</b>                                             | <p>(ii) Since <math>\angle GCD = \alpha</math> (shown above)<br/> <math>\angle GCB = 180^\circ - \alpha</math> (angles on a straight line)<br/> also, <math>\angle BFG = \alpha</math> (shown above)</p> <p><math>\angle GCB + \angle BFG = 180^\circ - \alpha + \alpha</math><br/> <math>= 180^\circ</math></p> <p><math>\therefore</math> BCGF is a cyclic quadrilateral (opposite angles supplementary)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p><b>2 marks</b><br/> Correct solution with full reasoning.<br/> <b>1 mark</b><br/> Substantial progress towards a correct solution.</p>                                                                            |
| <b>HE4</b>                                             | <p>(b) (i)</p> $\frac{d}{dx} x \sin 3x = x \cdot 3 \cos 3x + \sin 3x \cdot 1$ $= 3x \cos 3x + \sin 3x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>2 marks</b><br/> Correct application of product rule to find correct answer.<br/> <b>1 mark</b><br/> Demonstrates knowledge of product rule in making substantial progress to a full solution.</p>             |
| <b>HE4</b>                                             | <p>(ii)</p> <p>If <math>\frac{d}{dx} x \sin 3x = 3x \cos 3x + \sin 3x</math><br/> then <math>3x \cos 3x = \frac{d}{dx} x \sin 3x - \sin 3x</math><br/> <math>x \cos 3x = \frac{1}{3} \frac{d}{dx} x \sin 3x - \frac{1}{3} \sin 3x</math><br/> Integrating both sides,</p> $\int_0^{\frac{\pi}{6}} x \cos 3x dx = \frac{1}{3} \int_0^{\frac{\pi}{6}} \frac{d}{dx} x \sin 3x dx - \frac{1}{3} \int_0^{\frac{\pi}{6}} \sin 3x dx$ $= \frac{1}{3} \left[ x \sin 3x + \frac{1}{3} \cos 3x \right]_0^{\frac{\pi}{6}}$ $= \frac{1}{3} \left[ \left( \frac{\pi}{6} \sin \frac{\pi}{2} + \frac{1}{3} \cos \frac{\pi}{2} \right) - \left( 0 \sin 0 + \frac{1}{3} \cos 0 \right) \right]$ $= \frac{1}{3} \left[ \left( \frac{\pi}{6} - 0 \right) - \left( 0 + \frac{1}{3} \right) \right]$ $= \frac{1}{3} \left( \frac{\pi}{6} - \frac{1}{3} \right)$ $= \frac{\pi - 2}{18}$ | <p><b>3 marks</b><br/> Correct solution.<br/> <b>2 marks</b><br/> Correctly finds the required primitive function.<br/> <b>1 mark</b><br/> Substantial progress towards finding the required primitive function.</p> |



|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>HE6</b></p> | <p>(c)</p> <p>Let <math>y = \sqrt{x}</math><br/> <math>\therefore x = y^2</math><br/> <math>\frac{dx}{dy} = 2y</math><br/> <math>dx = 2ydy</math><br/> <math display="block">\int \frac{dx}{\sqrt{x(1-x)}} = \int \frac{2ydy}{\sqrt{y^2(1-y^2)}}</math><br/> <math display="block">= \int \frac{2ydy}{y\sqrt{(1-y^2)}}</math><br/> <math display="block">= 2 \int \frac{dy}{\sqrt{(1-y^2)}}</math><br/> <math display="block">= 2 \sin^{-1} y + c</math><br/> but <math>y = \sqrt{x}</math><br/> <math display="block">\therefore \int \frac{dx}{\sqrt{x(1-x)}} = 2 \sin^{-1} \sqrt{x} + c</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p><b>3 marks</b><br/>Correct solution.</p> <p><b>2 marks</b><br/>Uses the given substitution correctly and makes substantial progress towards a correct solution.</p> <p><b>1 mark</b><br/>Uses the given substitution correctly.</p>                                     |
| <p><b>HE2</b></p> | <p>(d)</p> <p><math>n! &gt; 2^n</math>, for all positive integral values of <math>n \geq 4</math><br/> Prove true for <math>n = 4</math><br/> <math>LHS = 4! \qquad RHS = 2^4</math><br/> <math>\qquad = 24 \qquad \qquad = 16</math><br/> <math>24 &gt; 16</math><br/> <math>\therefore</math> True for <math>n = 4</math></p> <p>Assume true for <math>n = k</math><br/> ie. Assume <math>k! &gt; 2^k</math><br/> <math>k! - 2^k &gt; 0</math></p> <p>Prove true for <math>n = k + 1</math><br/> ie. Prove <math>(k + 1)! &gt; 2^{k+1}</math><br/> Consider the difference<br/> <math display="block">(k + 1)! - 2^{k+1} = (k + 1)k! - 2 \cdot 2^k</math> <math display="block">= k \cdot k! + k! - 2^k - 2^k</math> <math display="block">= k \cdot k! - 2^k + k! - 2^k</math> <math display="block">= (k - 1)k! + k! - 2^k + k! - 2^k</math> <math display="block">= (k - 1)k! + 2(k! - 2^k)</math> <p>Now, since <math>k &gt; 4</math>, <math>(k - 1) &gt; 0</math><br/> <math>k! &gt; 0</math><br/> <math>\therefore (k - 1)k! &gt; 0</math><br/> Also, <math>k! - 2^k &gt; 0</math>, from the assumption<br/> Hence, <math>(k - 1)k! + 2(k! - 2^k) &gt; 0</math><br/> <math>(k + 1)! - 2^{k+1} &gt; 0</math><br/> <math>\therefore (k + 1)! &gt; 2^{k+1}</math></p> <p><math>\therefore</math> By the process of mathematical induction,<br/> <math>n! &gt; 2^n</math> is true for all positive integral values of <math>n \geq 4</math></p> </p> | <p><b>3 marks</b><br/>Correct solution.</p> <p><b>2 marks</b><br/>Prove the relationship is true for <math>n=4</math> and makes substantial progress towards a correct solution.</p> <p><b>1 mark</b><br/>Correctly proves the relationship true for <math>n=4</math>.</p> |